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Asymmetric introduction of steam into turbine engine combustion chamber

Abstract

A method is provided for operating a turbine engine. During this method, fuel is injected into a combustion chamber of a combustor of the turbine engine. The combustion chamber extends circumferentially around a centerline. The fuel is combusted within the combustion chamber. Steam is introduced into the combustion chamber in an asymmetric pattern around the centerline.

Inventors:	Boardman; Gregory (Owens Cross Roads, AL)
Applicant:	Raytheon Technologies Corporation (Farmington, CT)
Family ID:	1000008819914
Assignee:	RTX Corporation (Farmington, CT)
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Primary Examiner: Sung; Gerald L

Assistant Examiner: Lisowski; Jacek

Attorney, Agent or Firm: Getz Balich LLC

Background/Summary

BACKGROUND OF THE DISCLOSURE

1. Technical Field

(1) This disclosure relates generally to a turbine engine and, more particularly, to utilizing steam during operation of the turbine engine.

2. Background Information

(2) As government emissions standards tighten, interest in alternative fuels for gas turbine engines continues to grow. There is interest, for example, in fueling a gas turbine engine with hydrogen (H.sub.2) fuel rather than a traditional hydrocarbon fuel such as kerosene to reduce greenhouse emissions. Combustion products produced by combusting hydrogen (H.sub.2) fuel include water vapor. Various systems and methods are known in the art for recovering the water vapor. Various system and methods are also known in the art for producing and utilizing steam from the recovered water vapor. While these known systems and methods have various advantages, there is still room in the art for improvement.

SUMMARY OF THE DISCLOSURE

(3) According to an aspect of the present disclosure, a method is provided for operating a turbine engine. During this method, fuel is injected into a combustion chamber of a combustor of the turbine engine. The combustion chamber extends circumferentially around a centerline. The fuel is combusted within the combustion chamber. Steam is introduced into the combustion chamber in an asymmetric pattern around the centerline.

(4) According to another aspect of the present disclosure, an assembly is provided for a turbine engine. This assembly includes a combustor, a fuel system and a steam system. The combustor includes a combustion chamber and a centerline. The combustion chamber extends circumferentially around the centerline. The fuel system is configured to inject fuel into the combustion chamber for combustion. The steam system is configured to introduce steam into the combustion chamber in an asymmetric pattern around the centerline.

(5) According to still another aspect of the present disclosure, another assembly is provided for a turbine engine. This assembly includes a combustor, a fuel system and a steam system. The combustor includes a combustion chamber and a centerline. The combustion chamber extends circumferentially around the centerline. The fuel system is configured to inject fuel into the combustion chamber for combustion. The steam system is configured to introduce steam into the combustion chamber through at least a plurality of first orifices and a plurality of second orifices. The first orifices and the second orifices are arranged in a non-repeating pattern circumferentially around the centerline. The steam system is configured to introduce the steam into the combustion chamber through each of the first orifices at a first flowrate. The steam system is configured to introduce the steam into the combustion chamber through each of the second orifices at a second flowrate that is different than the first flowrate.

(6) The asymmetric pattern may be divided into a first half sector and a second half sector around the centerline. The steam system may be configured to introduce the steam into the combustion chamber in a first pattern about the centerline in the first half sector. The steam system may be configured to introduce the steam into the combustion chamber in a second pattern about the centerline in the second half sector. The second pattern may be different than the first pattern.

(7) The asymmetric pattern may be divided into a first half sector and a second half sector around the centerline. The steam system may be configured to introduce a first quantity of the steam into the combustion chamber in the first half sector. The steam system may be configured to introduce a second quantity of the steam into the combustion chamber in the second half sector. The second quantity of the steam may be different than the first quantity of the steam.

(8) The method may also include attenuating acoustic modes generated by the combustion of the fuel using the steam introduced into the combustion chamber in the asymmetric pattern.

(9) The asymmetric pattern may be divided into N number of sectors around the centerline with equal arc lengths. The N number of sectors may include a first sector and a second sector. The steam may be introduced into the combustion chamber in a first pattern about the centerline in the first sector. The steam may be introduced into the combustion chamber in a second pattern about

- the centerline in the second sector. The second pattern may be different than the first pattern.
- (10) A first quantity of the steam may be introduced into the combustion chamber in the first sector. A second quantity of the steam may be introduced into the combustion chamber in the second sector. The second quantity of the steam may be different than the first quantity of the steam.
- (11) The N number may be an even number.
- (12) The N number may be an odd number.
- (13) The asymmetric pattern may be divided into N number of sectors around the centerline with equal arc lengths. The N number of sectors may include a first sector and a second sector. A first quantity of the steam may be introduced into the combustion chamber in the first sector. A second quantity of the steam may be introduced into the combustion chamber in the second sector. The second quantity of the steam may be different than the first quantity of the steam.
- (14) The steam may be introduced into the combustion chamber in the asymmetric pattern at a plurality of steam injection locations. The steam injection locations may be arranged in an array and equally spaced around the centerline.
- (15) The steam may be introduced into the combustion chamber in the asymmetric pattern through a plurality of orifices with a common size.
- (16) The steam may be introduced into the combustion chamber in the asymmetric pattern through a first orifice and a second orifice with a different size than the first orifice.
- (17) The steam may be introduced into the combustion chamber in the asymmetric pattern along at a bulkhead of the combustor.
- (18) The steam may be introduced into the combustion chamber through a plurality of fuel injector assemblies. The fuel may be injected into the combustion chamber through the plurality of fuel injector assemblies.
- (19) The steam may be introduced into the combustion chamber in the asymmetric pattern along at a wall of the combustor.
- (20) The steam may be introduced into the combustion chamber in the asymmetric pattern through a plurality of quench apertures.
- (21) The fuel may be injected into the combustion chamber in a symmetric pattern around the centerline.
- (22) The method may also include changing the asymmetric pattern based on an operating mode of the turbine engine.
- (23) The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.
- (24) The foregoing features and the operation of the invention will become more apparent in light of the following description and the accompanying drawings.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a partial schematic illustration of a gas turbine engine.
- (2) FIG. 2 is a partial schematic illustration of a combustor section between a compressor section and a turbine section.
- (3) FIG. 3 is a perspective illustration of a combustor.
- (4) FIG. 4 is a partial sectional illustration of a combustor wall.
- (5) FIG. 5 is a schematic illustration of a fuel system configured with the combustor section.
- (6) FIG. 6 is a schematic illustration of a steam system configured with the combustor section.
- (7) FIG. 7 is a schematic illustration of a portion of the steam system configured with the combustor section, where the steam system has various outlet orifice sizes.
- (8) FIG. 8 is a partial schematic illustration of a portion of the gas turbine engine configured with a

steam delivery circuit.

(9) FIG. 9 is a flow diagram of a method for operating a gas turbine engine.

DETAILED DESCRIPTION

(10) FIG. 1 is a side sectional illustration of a gas turbine engine 20 for an aircraft propulsion system. This turbine engine 20 extends axially along an axial centerline 22 between a forward, upstream end 24 and an aft, downstream end 26. The turbine engine 20 includes a fan section 28, a compressor section 29, a combustor section 30 and a turbine section 31. The compressor section 29 of FIG. 1 includes a low pressure compressor (LPC) section 29A and a high pressure compressor (HPC) section 29B. The turbine section 31 of FIG. 1 includes a high pressure turbine (HPT) section 31A and a low pressure turbine (LPT) section 31B.

(11) The engine sections 28-31B of FIG. 1 are arranged sequentially along the axial centerline 22 within an engine housing 32. This engine housing 32 includes an inner case 34 (e.g., a core case) and an outer case 36 (e.g., a fan case). The inner case 34 may house one or more of the engine sections 29A-31B; e.g., a core of the turbine engine 20. The outer case 36 may house at least the fan section 28.

(12) Each of the engine sections 28, 29A, 29B, 31A and 31B includes a respective bladed rotor 38-42. Each of these bladed rotors 38-42 includes a plurality of rotor blades arranged circumferentially around and connected to one or more respective rotor disks and/or hubs. The rotor blades, for example, may be formed integral with or mechanically fastened, welded, brazed, adhered and/or otherwise attached to the respective rotor disk(s) and/or the respective hub(s).

(13) The fan rotor 38 is connected to a geartrain 44, for example, through a fan shaft 46. The geartrain 44 and the LPC rotor 39 are connected to and driven by the LPT rotor 42 through a low speed shaft 47. The HPC rotor 40 is connected to and driven by the HPT rotor 41 through a high speed shaft 48. The engine shafts 46-48 are rotatably supported by a plurality of bearings; e.g., rolling element and/or thrust bearings. Each of these bearings is connected to the engine housing 32 by at least one stationary structure such as, for example, an annular support strut.

(14) During engine operation, air enters the turbine engine 20 through an airflow inlet 50 into the turbine engine 20. This air is directed through the fan section 28 and into a core flowpath 52 and a bypass flowpath 54. The core flowpath 52 extends sequentially through the engine sections 29A-31B (e.g., the engine core) from an inlet 56 into the core flowpath 52 to an exhaust 58 from the core flowpath 52. The air within the core flowpath 52 may be referred to as "core air". The bypass flowpath 54 extends through a bypass duct, and bypasses the engine core. The air within the bypass flowpath 54 may be referred to as "bypass air".

(15) The core air is compressed by the LPC rotor 39 and the HPC rotor 40 and directed into a (e.g., annular) combustion chamber 60 of a (e.g., annular) combustor 62 in the combustor section 30. Fuel is injected by one or more fuel injector assemblies 64 (one visible in FIG. 1) into the combustion chamber 60 and mixed with the compressed core air to provide a fuel-air mixture. This fuel-air mixture is ignited and combustion products thereof flow through and sequentially cause the HPT rotor 41 and the LPT rotor 42 to rotate before being directed out of the turbine engine 20 through the core exhaust 58. The rotation of the HPT rotor 41 and the LPT rotor 42 respectively drive rotation of the HPC rotor 40 and the LPC rotor 39 and, thus, compression of the air received from the core inlet 56. The rotation of the LPT rotor 42 also drives rotation of the fan rotor 38, which propels the bypass air through the bypass flowpath 54 and out of the turbine engine 20 through an exhaust 66 from the bypass flowpath 54. The propulsion of the bypass air may account for a majority of thrust generated by the turbine engine 20.

(16) FIG. 2 illustrate a portion of the combustor section 30 along the core flowpath 52 between the HPC section 29B and the HPT section 31A. This combustor section 30 includes the combustor 62, a diffuser plenum 68 and the one or more injector assemblies 64 (one visible in FIG. 2). Briefly, the combustor 62 is disposed within (e.g., surrounded by) the diffuser plenum 68. This diffuser plenum 68 receives the compressed core air from the HPC section 29B for subsequent provision into the

combustion chamber **60**. Each injector assembly **64** of FIG. 2 includes a fuel injector **70** mated with an air swirler structure **72**. The fuel injector **70** injects the fuel into the combustion chamber **60**. The air swirler structure **72** directs some of the core air from the diffuser plenum **68** into the combustion chamber **60** in a manner that facilitates mixing the core air with the injected fuel. One or more igniters (not shown) ignite the fuel-air mixture within the combustion chamber **60**. One or more quench apertures **74A**, **74B** (generally referred to as “**74**”) (e.g., dilution holes) in each wall **76A**, **76B** (generally referred to as “**76**”) of the combustor **62** direct additional core air from the diffuser plenum **68** into the combustion chamber **60** to quench (e.g., stoichiometrically lean) the combustion products; e.g., the ignited fuel-air mixture.

(17) The combustor **62** may be configured as an annular combustor; e.g., an annular floating wall combustor. The combustor **62** of FIGS. 2 and 3, for example, includes an annular combustor bulkhead **78**, the tubular inner combustor wall **76A** (“inner wall”), and the tubular outer combustor wall **76B** (“outer wall”). The bulkhead **78** of FIG. 2 extends radially between and to the inner wall **76A** and the outer wall **76B**. The bulkhead **78** may be connected (e.g., mechanically fastened or otherwise attached) to the inner wall **76A** and/or the outer wall **76B**. Each combustor wall **76** projects axially along the axial centerline **22** out from the bulkhead **78** towards the HPT section **31A**. The inner wall **76A** of FIG. 2, for example, projects axially to and may be connected to an inner platform **80A** of a downstream stator vane array **82** (e.g., a turbine inlet nozzle) in the HPT section **31A**. The outer wall **76B** of FIG. 2 projects axially to and may be connected to an outer platform **80B** of the downstream stator vane array **82**. With the arrangement of FIG. 2, the combustion chamber **60** is formed by and extends radially within the combustor **62** between and to the inner wall **76A** and the outer wall **76B**. The combustion chamber **60** is formed by and extends axially (in an upstream direction along the core flowpath **52**) into the combustor **62** from the stator vane array **82** to the bulkhead **78**. The combustion chamber **60** also extends within the combustor **62** circumferentially about (e.g., completely around) the axial centerline **22**, which may configure the combustion chamber **60** as a full-hoop annulus.

(18) Any one or more or all of the walls **76A**, **76B**, **78** may each be configured as a multi-walled structure; e.g., a hollow, dual-walled structure. For example, referring to FIG. 4, each wall **76A**, **76B**, **78** includes a combustor wall shell **84**, a combustor wall heat shield **86** (e.g., a liner) and one or more combustor wall cooling cavities **88** (e.g., impingement cavities) formed by and (e.g., radially and/or axially) between the shell **84** and the heat shield **86**. Each cooling cavity **88** of FIG. 4 is fluidly coupled with the diffuser plenum **68** through one or more cooling apertures **90** in the shell **84**; e.g., impingement apertures. Each cooling cavity **88** of FIG. 4 is fluidly coupled with the combustion chamber **60** through one or more cooling apertures **92** in the heat shield **86**; e.g., effusion apertures. Of course, various other multi-walled combustor wall structures are known in the art, and the present disclosure is not limited to any particular ones thereof. Furthermore, it is contemplated any one or more or all of the walls **76A**, **76B** and/or **78** of FIG. 2 may each alternatively be configured as a single-walled structure. The shell **84** of FIG. 4, for example, may be omitted and the heat shield **86** may form a single walled liner/wall. However, for ease of description, each wall **76A**, **76B**, **78** may each be described below as the hollow, dual-walled structure.

(19) Referring to FIG. 2, the stator vane array **82** includes the inner platform **80A**, the outer platform **80B** and a plurality of stator vanes **94** (one visible in FIG. 2); e.g., hollow stator vanes. The stator vanes **94** are arranged circumferentially about the axial centerline **22** in an array; e.g., a circular array. Each of these stator vanes **94** extends radially across the core flowpath **52** between and to the inner platform **80A** and the outer platform **80B**. Each of the stator vanes **94** may also be connected to the inner platform **80A** and/or the outer platform **80B**. The stator vane array **82** and its stator vanes **94** are configured to turn and/or otherwise condition the combustion products exiting the combustion chamber **60** for interaction with a first stage of the HPT rotor **41** (see FIG. 1).

(20) Referring to FIG. 5, the turbine engine **20** includes a fuel system **96** for delivering the fuel to

the combustor **62**. This fuel system **96** includes a fuel source **98** and the one or more fuel injectors **70**. The fuel source **98** of FIG. **5** includes a fuel reservoir **100** and/or a fuel flow regulator **102**; e.g., a valve and/or a pump. The fuel reservoir **100** is configured to store the fuel before, during and/or after turbine engine operation. The fuel reservoir **100**, for example, may be configured as or otherwise include a tank, a cylinder, a pressure vessel, a bladder or any other type of fuel storage container. The fuel flow regulator **102** is configured to direct and/or meter a flow of the fuel from the fuel reservoir **100** to one or more or all of the fuel injectors **70**. The fuel injectors **70** may be arranged circumferentially about the axial centerline **22** in an array. Each fuel injector **70** is configured to direct the fuel received from the fuel source **98** into the combustion chamber **60** for combustion.

(21) The fuel delivered by the fuel system **96** may be a non-hydrocarbon fuel; e.g., a hydrocarbon free fuel. Examples of the non-hydrocarbon fuel include, but are not limited to, hydrogen fuel (e.g., hydrogen (H.sub.2) gas) and ammonia fuel (e.g., ammonia (NH.sub.3) gas). The turbine engine **20** of FIG. **1** may thereby be configured as a non-hydrocarbon turbine engine; e.g., a hydrocarbon free turbine engine. The present disclosure, however, is not limited to non-hydrocarbon turbine engines. The fuel delivered by the fuel system **96**, for example, may alternatively be a hydrocarbon fuel such as, but not limited to, kerosene or jet fuel. The turbine engine **20** of FIG. **1** may thereby be configured as a hydrocarbon turbine engine. Alternatively, the fuel system **96** may be configured as a multi-fuel system operable to deliver, individually or in combination, multiple different fuels (e.g., a non-hydrocarbon fuel and a hydrocarbon fuel, etc.) for combustion within the combustion chamber **60**. The turbine engine **20** of FIG. **1** may thereby be configured as a multi-fuel turbine engine; e.g., a dual-fuel turbine engine. However, for ease of description, the fuel delivered by the fuel system **96** may be described below as the non-hydrocarbon fuel; e.g., the hydrogen fuel.

(22) Combustion of the fuel-air mixture within the combustion chamber **60** may generate acoustic modes (e.g., acoustic-heat-release-coupling modes) within the combustion products during various modes (e.g., power settings) of turbine engine operation. These acoustic modes may increase engine noise, increase engine vibrations, create combustion instabilities and/or reduce combustion efficiency. Referring to FIG. **6**, to attenuate (e.g., reduce, eliminate, etc.) or otherwise disrupt some or all of these acoustic modes, the turbine engine **20** includes a steam system **104**. This steam system **104** is configured to selectively introduce steam into the combustion chamber **60** to interact with the fuel-air mixture and/or the combustion products. The steam system **104** of FIG. **6** includes a steam source **106**, a steam delivery circuit **108** and a plurality of steam delivery devices **110A**, **110B** and **110C** (generally referred to as “**110**”).

(23) The steam source **106** is configured to provide the steam to the steam delivery circuit **108** during turbine engine operation and, more particularly, during steam system operation. The steam source **106**, for example, may be configured as or otherwise include an evaporator **112**, which may be or otherwise include a fluid-to-fluid heat exchanger and/or an electrical heater. The evaporator **112** is configured to evaporate water into the steam during the steam system operation. The water may be received from various sources. The steam source **106** of FIG. **6**, for example, includes a water reservoir **114** fluidly coupled with and upstream of the evaporator **112**. This water reservoir **114** is configured to store the water before, during and/or after turbine engine operation. Examples of the water reservoir **114** include, but are not limited to, a tank, a cylinder, a pressure vessel, a bladder or any other type of water storage container. Briefly, the water may be supplied to the water reservoir **114** by recovering water vapor from the combustion products flowing through the core flowpath **52** (see FIG. **1**) and/or from another water source onboard or offboard an aircraft.

(24) The steam delivery circuit **108** of FIG. **6** includes a supply circuit **116** and a plurality of feed circuits **118A**, **118B** and **118C** (generally referred to as “**118**”), where each of the feed circuits **118A**, **118B**, **118C** is associated with a respective one of the steam delivery devices **110A**, **110B**, **110C**. The supply circuit **116** of FIG. **6** extends from an outlet from the steam source **106** to an interface **120** with the feed circuits **118** such as a manifold. At this interface **120**, the feed circuits

118 may be fluidly coupled in parallel to and downstream of the supply circuit **116**. Each of the feed circuits **118** extends from the interface **120** to an inlet of a respective one of the steam delivery devices **110**.

(25) Each steam delivery device **110** is configured to introduce (e.g., direct) a portion of the steam directly or indirectly into the combustion chamber **60**. Examples of the steam delivery devices **110** include, but are not limited to, nozzles, diffusers, showerheads or any other devices operable to direct (e.g., inject, exhaust and/or otherwise flow) the steam along predetermined trajectories and/or into predetermined spaces. One or more of all of these steam delivery devices **110** may each be configured with (e.g., integrated into) another component of the turbine engine **20**. Alternatively, one or more or all of the steam delivery devices **110** may each be configured as a discrete (e.g., standalone) component of the turbine engine **20**.

(26) The steam system **104** of FIG. **6** is configured to introduce the steam into the combustion chamber **60** through/using at least or only the steam delivery devices **110** in an asymmetric pattern around the axial centerline **22**. For example, the steam system **104** of FIG. **6** is configured to direct the steam at a first flowrate into the combustion chamber **60** through each first steam delivery device **110A**. The steam system **104** of FIG. **6** is configured to direct the steam at a second flowrate into the combustion chamber **60** through each second steam delivery device **110B**, where the second flowrate is different (e.g., greater) than the first flowrate. The steam system **104** of FIG. **6** is configured to direct the steam at a third flowrate into the combustion chamber **60** through each third steam delivery device **110C**, where the third flowrate is different (e.g., greater) than the first flowrate and/or the second flowrate. These steam delivery devices **110A**, **110B** and **110C** are interspersed with one another and arranged in an annular (e.g., circular) array about the axial centerline **22**. Within the delivery device array, the steam delivery devices **110** may be arranged in a non-repeating pattern around the axial centerline **22**.

(27) The asymmetric pattern of FIG. **6** may be divided into N number of sectors (e.g., **122A** and **122B**; generally referred to as “**122**”) around the axial centerline **22**. Here, the number N is an even number such as, but not limited to, two (2), four (4), six (6), etc. For example, the asymmetric pattern of FIG. **6** is divided into half sectors **122A** and **122B**. However, the number N may alternatively be an odd number such as, but not limited to, three (3), five (5), seven (7), etc. For example, the asymmetric pattern may be divided into third sectors.

(28) Each of the pattern sectors **122** of FIG. **2** is circumferentially between and may be abutted against (e.g., directly next to) opposing circumferential sides of the other pattern sector **122**, or a circumferentially neighboring (e.g., adjacent) pair of the pattern sectors **122** where N is equal to or greater than three (3). The pattern sectors **122** may have common (e.g., the same, equal) arc lengths, where the arc length is measured about the axial centerline **22** between opposing circumferential sides of the respective pattern sector **122**. However, one or more or all of the pattern sectors **122** may each have a unique steam flow pattern. For example, the steam is introduced into the combustion chamber **60** in a first pattern about the axial centerline **22** in the first pattern sector **122A**. The steam is introduced into the combustion chamber **60** in a second pattern about the axial centerline **22** in the second pattern sector **122B**, where the second pattern is different than the first pattern. For example, the arrangement of the steam delivery devices **110** (e.g., a sequence of **110C-110C-110A-110A-110A-110C-110C-110A** in a clockwise direction) within the first pattern sector **122A** is different than the arrangement of the steam delivery devices **110** (e.g., a sequence of **110A-110B-110C-110B-110A-110A-110B-110A** in a clockwise direction) within the second pattern sector **122B**. In addition or alternatively, the steam delivery devices **110** within the first pattern sector **122A** may be configured to direct a first quantity of the steam into the combustion chamber **60**. By contrast, the steam delivery devices **110** within the second pattern sector **122B** may be configured to direct a second quantity of the steam into the combustion chamber **60** that is different (e.g., greater or less) than the first quantity of the steam. With such an arrangement, the asymmetric pattern of the steam introduced into the combustion chamber **60** may

attenuate or otherwise disrupt some or all of the acoustic modes in the combustion products. Circumferential locations of the steam delivery devices **110** may be equispaced (or not) about the axial centerline **22**.

(29) The flow of the steam through the steam delivery devices **110** may be tailored by regulating the flow to the steam delivery devices **110**. For example, each feed circuit **118A**, **118B**, **118C** of FIG. **6** is configured with a steam flow regulator **124A**, **124B**, **124C** (generally referred to as “**124**”), only some of which are labeled in FIG. **6** for ease of illustration. Each steam flow regulator **124** is arranged (e.g., fluidly coupled inline) with a respective one of the steam delivery devices **110**. Each steam flow regulator **124** is configured to selectively direct and/or meter a flow of the steam from the steam source **106** to a respective one of the steam delivery devices **110**. For example, each steam flow regulator **124** may be configured as or otherwise include a control valve. This control valve may fully open, may fully close and/or may move to one or more partially open positions. While each steam flow regulator **124** is illustrated in FIG. **6** as being part of the respective feed circuit **118**, that steam flow regulator **124** may alternatively be arranged at the interface **120** between the supply circuit **116** and the respective feed circuit **118**, at the outlet from the respective feed circuit **118**, or otherwise. Furthermore, it is contemplated a single steam flow regulator **124A**, **124B**, **124C** may regulate the flow of the steam to like steam delivery devices **110A**, **110B**, **110C**.

(30) Referring to FIG. **7**, the flow of the steam through the steam delivery devices **110** may also or alternatively be tailored by selectively sizing the steam delivery devices **110**. For example, a size of an output orifice from each first steam delivery device **110A** may be different (e.g., smaller) than a size of an output orifice from each second steam delivery device **110B**. Similarly, the size of the output orifice from each second steam delivery device **110B** may be different (e.g., smaller) than a size of an output orifice from each third steam delivery device **110C**. With such an arrangement, the steam through the steam delivery devices **110** may be tailored without use of multiple steam flow regulators **124**. Of course, in other embodiments, some or all of the steam delivery devices **110** may alternatively have a common size.

(31) In some embodiments, referring to FIG. **8**, one or more or all of the steam delivery devices **110** (labeled as “**110**” in FIG. **8**) may each be configured as part of a respective one of the injector assemblies **64**. The steam delivery device **110'** of FIG. **8** includes an internal steam passage **126** extending within the fuel injector **70** to at least one steam outlet orifice **128** at a tip of the fuel injector **70**. Here, the steam delivery device **110'** is configured to direct (e.g., inject) the steam out of the steam outlet orifice **128**, through an inner bore of the air swirler structure **72**, and into the combustion chamber **60** at the bulkhead **78**. The steam delivery device **110'** of FIG. **8** may also (or alternatively) include an internal fuel passage **130** extending within the fuel injector **70** to at least one fuel outlet orifice **132** at the injector tip. Here, the injector assembly **64**/the steam delivery device **110'** is configured to direct a mixture of the fuel and the steam out of the fuel outlet orifice **132**, through the swirler bore, and into the combustion chamber **60** at the bulkhead **78**.

(32) In some embodiments, one or more or all of the steam delivery devices **110** (labeled as “**110**” in FIG. **8**) may each be configured to direct the steam into a volume **134** between the bulkhead **78** and a combustor hood **136**. This steam may flow into the combustion chamber **60** by flowing across the bulkhead **78** via its cooling apertures and/or cooling cavities. The steam may also or alternatively flow into the combustion chamber **60** by flowing through (e.g., being directed into) the air swirler structures **72**.

(33) In some embodiments, one or more or all of the steam delivery devices **110** (labeled as “**110**” in FIG. **8**) may each be configured to direct the steam into the diffuser plenum **68** adjacent the inner wall **76A**. This steam may flow into the combustion chamber **60** by flowing across the inner wall **76A** via its cooling apertures and/or cooling cavities. The steam may also or alternatively flow into the combustion chamber **60** by flowing through (e.g., being directed into) the inner quench apertures **74A**.

(34) In some embodiments, one or more or all of the steam delivery devices **110** (labeled as “**110**” in FIG. **8**) may each be configured to direct the steam into the diffuser plenum **68** adjacent the outer wall **76B**. This steam may flow into the combustion chamber **60** by flowing across the outer wall **76B** via its cooling apertures and/or cooling cavities. The steam may also or alternatively flow into the combustion chamber **60** by flowing through (e.g., being directed into) the outer quench apertures **74B**.

(35) The steam system **104** of FIG. **8** is shown with multiple arrays of the steam delivery devices **110'**, **110''**, **110'''**, **110''''**. Each of these arrays may be configured to introduce the steam into the combustion chamber **60** according to its own asymmetric pattern around the axial centerline **22**. These patterns may be the same or different. Furthermore, it is contemplated the any one or more of the multiple arrays may be omitted such that the steam system **104** includes any one, two or three of the arrays shown in FIG. **8**.

(36) The asymmetric pattern(s) are described above as being circumferentially extending patterns. It is contemplated, however, that the steam may also be introduced into the combustion chamber **60** in an axially asymmetric (or symmetric) pattern. Furthermore, it is also contemplated the fuel injected by the fuel injectors **70** and/or the quench air injected by the quench apertures **74** may be introduced into the combustion chamber **60** according to a respective asymmetric pattern or a symmetric pattern.

(37) The steam delivery devices **110**/arrays of steam delivery devices are described above for use in attenuation of the acoustic mode(s) in the combustion products. However, any one or more or all of these steam delivery devices **110**/arrays of steam delivery devices may also (or alternatively in certain modes of operation) be utilized for cooling a respective structure of the turbine engine **20**.

(38) FIG. **9** is a flow diagram of a method **900** for operating a gas turbine engine. For ease of description, the operating method **900** is described with respect to the turbine engine **20** of FIG. **1**, the fuel system **96** of FIG. **5** and the steam system **104** of FIG. **6**. The operating method **900** of the present disclosure, however, is not limited to such an exemplary arrangement.

(39) In step **902**, the fuel is injected into the combustion chamber **60** through the injector assemblies **64** and their fuel injectors **70**.

(40) In step **904**, the fuel is ignited within the combustion chamber **60** such that the fuel-air mixture combusts.

(41) In step **906**, the steam is introduced into the combustion chamber **60** in an asymmetric pattern. This asymmetric pattern may be tailored to attenuate the acoustic modes generated by the combustion of the fuel-air mixture within the combustion chamber **60**. The asymmetric pattern may also be tailored to cool one or more structures along and/or proximate the combustion chamber **60**.

(42) In some embodiments, the asymmetric pattern for introducing the steam into the combustion chamber **60** may remain constant across multiple modes of engine operation; e.g., across multiple engine power settings. However, in other embodiments, the asymmetric pattern for introducing the steam into the combustion chamber **60** may changes across multiple modes of engine operation; e.g., across multiple engine power settings. This may be particularly useful where the acoustic modes also change across the multiple modes of engine operation.

(43) The steam system **104** may be included in various turbine engines other than the one described above. The steam system **104**, for example, may be included in a geared turbine engine where a geartrain connects one or more shafts to one or more rotors in a fan section, a compressor section and/or any other engine section. Alternatively, the steam system **104** may be included in a turbine engine configured without a geartrain; e.g., a direct drive turbine engine. The steam system **104** may be included in a geared or non-geared turbine engine configured with a single spool, with two spools (e.g., see FIG. **1**), or with more than two spools. The turbine engine may be configured as a turbofan engine, a turbojet engine, a turboprop engine, a turboshaft engine, a propfan engine, a pusher fan engine or any other type of turbine engine. The turbine engine may alternatively be configured as an auxiliary power unit (APU) or an industrial gas turbine engine. The present

disclosure therefore is not limited to any particular types or configurations of turbine engines.

(44) While various embodiments of the present disclosure have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the disclosure. Accordingly, the present disclosure is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. A method for operating a turbine engine, comprising: injecting fuel into a combustion chamber of a combustor of the turbine engine, the fuel injected into the combustion chamber through a plurality of fuel injector assemblies, wherein the combustion chamber is an annulus that extends circumferentially around a centerline; combusting the fuel within the combustion chamber; and introducing steam into the combustion chamber in an asymmetric pattern around the centerline using a plurality of steam delivery devices, the steam introduced into the combustion chamber through respective tips of the plurality of fuel injector assemblies, wherein the asymmetric pattern is divided into N number of sectors around a circumference of the centerline with equal arc lengths, the N number of sectors includes a first sector and a second sector, the first sector includes a first of the plurality of steam delivery devices, the second sector includes a second of the plurality of steam delivery devices, the steam is introduced into the combustion chamber in a first pattern about the centerline in the first sector, and the steam is introduced into the combustion chamber in a second pattern about the centerline in the second sector, and the second pattern is different than the first pattern; wherein a first quantity of the steam is introduced into the combustion chamber in the first sector, and a second quantity of the steam is introduced into the combustion chamber in the second sector, and the second quantity of the steam is different than the first quantity of the steam.
2. The method of claim 1, further comprising attenuating acoustic modes generated by the combustion of the fuel using the steam introduced into the combustion chamber in the asymmetric pattern.
3. The method of claim 1, wherein the N number is an even number.
4. The method of claim 1, wherein the N number is an odd number.
5. The method of claim 1, wherein the plurality of steam delivery devices are arranged in an array and equally spaced around the centerline.
6. The method of claim 1, wherein the steam is introduced into the combustion chamber in the asymmetric pattern through a first orifice and a second orifice with a different size than the first orifice.
7. The method of claim 1, wherein the steam is introduced into the combustion chamber in the asymmetric pattern along a bulkhead of the combustor.
8. The method of claim 1, wherein the fuel is injected into the combustion chamber in a symmetric pattern around the centerline.
9. The method of claim 1, further comprising changing the asymmetric pattern based on an operating mode of the turbine engine.
10. An assembly for a turbine engine, comprising: a combustor comprising a combustion chamber and a centerline, wherein the combustion chamber is an annulus that extends circumferentially around the centerline; a fuel system configured to inject fuel into the combustion chamber for combustion; and a steam system configured to introduce steam into the combustion chamber in an asymmetric pattern around the annulus using a plurality of steam delivery devices, the steam system comprising a steam source, a steam delivery circuit, and a plurality of steam delivery

devices, wherein the asymmetric pattern is divided into a first half sector and a second half sector around the centerline, the steam system is configured to introduce the steam into the combustion chamber in a first pattern about the centerline in the first half sector, the steam system is configured to introduce the steam into the combustion chamber in a second pattern about the centerline in the second half sector, and the second pattern is different than the first pattern; the steam delivery circuit including a supply circuit and a plurality of feed circuits, the supply circuit extending from the steam source to an interface of the plurality of feed circuits, each of the plurality of feed circuits extending from the interface to a respective one of the plurality of steam delivery devices.

11. The assembly of claim 10, wherein the steam system is configured to introduce a first quantity of the steam into the combustion chamber in the first half sector; and the steam system is configured to introduce a second quantity of the steam into the combustion chamber in the second half sector, and the second quantity of the steam is different than the first quantity of the steam.

12. An assembly for a turbine engine, comprising: a combustor comprising an annular combustion chamber and a centerline, the annular combustion chamber extending circumferentially around the centerline; a fuel system configured to inject fuel into the annular combustion chamber for combustion; and a steam system configured to introduce steam into the annular combustion chamber through at least a plurality of first orifices and a plurality of second orifices, the steam system including a first feed circuit, a second feed circuit and a supply circuit, the first feed circuit in communication with the plurality of first orifices, the second feed circuit in communication with the plurality of second orifices, the supply circuit in parallel fluid communication with the first feed circuit and the second feed circuit, the plurality of first orifices and the plurality of second orifices arranged in a non-repeating pattern circumferentially around the centerline, the steam system configured to introduce the steam into the annular combustion chamber through each of the plurality of first orifices at a first flowrate, and the steam system configured to introduce the steam into the annular combustion chamber through each of the plurality of second orifices at a second flowrate that is different than the first flowrate.
